Magnetometer-Free 6DOF Sensor Fusion: Overcoming Yaw Drift and Geometric Singularity in 6-Axis MEMS IMUs for Humanoid Teleoperation

Abstract

Immersive teleoperation of highly articulated humanoid robots demands low-latency, high-fidelity kinematic tracking. While wearable Inertial Measurement Unit (IMU) harnesses are the industry standard for upper-body motion capture, traditional 9-axis systems rely on magnetometers that are highly susceptible to the severe electromagnetic interference generated by humanoid servos and laboratory environments. Utilizing a 6-axis MEMS sensor (MPU6050) eliminates magnetic vulnerability but introduces catastrophic unreferenced yaw drift and geometric cross-axis projection errors during unconstrained 3D spatial movement. This thesis presents a novel asynchronous firmware architecture and mathematical sensor fusion pipeline designed to achieve magnetometer-level yaw stability utilizing strictly 6-axis hardware. By uncapping silicon measurement ceilings to ±2000 °/s, decoupling kinematic mathematics into an uninterrupted 500Hz hardware loop, and implementing a rigorously calibrated Madgwick AHRS gradient descent filter, the system transitions from fragile 1D Euler tracking to continuous 4-dimensional quaternion integration. The resulting architecture successfully neutralizes Gimbal Lock, network-induced Nyquist failures, and high-G linear distortion, reducing kinematic yaw drift from 24 degrees per 30 seconds to less than 1 degree over extended periods of unconstrained, high-velocity 3D motion.

1. Introduction

The seamless control of humanoid robots, such as the Unitree G1, requires a teleoperation framework capable of mirroring human anatomical complexity in real-time. Wearable motion-capture harnesses utilizing micro-electromechanical systems (MEMS) are favored for their low cost, high update rates, and untethered form factors. However, translating the chaotic, high-acceleration dynamics of a human arm into continuous, drift-free joint angles presents a significant mathematical and hardware engineering challenge.

The prevailing methodology for spatial orientation tracking relies on 9-axis IMUs, which fuse an accelerometer, gyroscope, and magnetometer to establish a 9-Degrees-of-Freedom (9DOF) coordinate system. While the magnetometer provides a critical absolute reference to Earth's magnetic north to prevent yaw drift, it introduces a fatal flaw in robotics applications: profound vulnerability to electromagnetic interference. The high-draw power supplies, metal infrastructure, and massive electric motors inherent to humanoid operation constantly warp the local magnetic field, rendering compass-based yaw tracking volatile and unusable in close proximity to the robot.

Isolating the teleoperation harness from magnetic interference requires downgrading to a 6-axis IMU (accelerometer and gyroscope). This solves the environmental vulnerability but leaves the horizontal yaw axis entirely unreferenced by global gravity. Without mathematical intervention, raw 6-axis gyroscope integration rapidly succumbs to static hardware bias, high-G measurement clipping, and geometric projection errors, resulting in permanent spatial disorientation. Furthermore, mapping 3D arm movements using standard Euler angles (Yaw,

Pitch, Roll) inevitably subjects the kinematic stream to Gimbal Lock—a mathematical singularity that destroys spatial tracking when anatomical joints rotate past 90 degrees.

This research addresses these fundamental limitations by restructuring the IMU firmware architecture at the metal level. By synthesizing 500Hz asynchronous network protocols, hardware-level Digital Signal Processor (DSP) flushing, and continuous 4D quaternion sensor fusion, this methodology aims to construct a highly resilient 6DOF tracking pipeline. The primary objective is to force a standard 6-axis sensor to punch above its hardware class, achieving the yaw stability of a 9-axis system entirely through optimized kinematics.

2. Planar (1D) Kinematics and Hardware Baseline Optimization

Before introducing the complex geometric projections required for unconstrained 3D space, the initial diagnostic phase isolated the IMU on a flat, 2D planar surface. By restricting physical movement strictly to the Z-axis (Yaw), gravitational influence was neutralized on the X and Y axes. This 1D planar environment was critical for isolating fundamental hardware flaws, signal-to-noise ratios, and micro-controller processing limitations without the confounding variables of 3D sensor fusion.

The objective of this phase was to establish a flawless, single-axis kinematic baseline by integrating angular velocity (ω) over time (Δt) to derive angular position (θ).

2.1 Iteration I: Raw Riemann Integration and the Silicon Saturation Ceiling

Methodology:

The foundational tracking algorithm utilized a standard Riemann sum numerical integration. As the sensor was rotated along the planar Z-axis, the raw angular velocity from the gyroscope (ω_z) was multiplied by the time elapsed since the previous loop iteration (Δt). The resulting discrete angular displacement was continuously added to a global heading variable:

$$\theta_t = \theta_{t-1} + (\omega_z \times \Delta t)$$

The initial trial consisted of a 30-second stress test involving rapid, bidirectional, high-acceleration manual rotations (mimicking aggressive VR arm swings).

Findings:

The raw integration failed severely, yielding a permanent, unrecoverable drift of 24 degrees at the conclusion of the 30-second trial. Telemetry analysis of the raw gyroscope output revealed a severe mechanical data anomaly: asymmetric signal clipping. The default firmware configuration of the MPU6050's GYRO_CONFIG register limits the maximum detectable rotation to $\pm\ 500^{\circ}/s$. During the violent apex of the physical arm swings, actual velocities routinely exceeded 800$^\circ/s$. Because the hardware was structurally "blind" to any velocity above 500$^\circ/s$, the mathematical area under the curve was artificially truncated at the peaks, permanently corrupting the orientation state.

Results & Resolution:
To eliminate hardware saturation, the MPU6050's internal restrictor plates were removed. The initialization sequence was rewritten to configure the gyroscope to its absolute maximum operational threshold of MPU6050_RANGE_2000_DEG ($\pm\ 2000^{\circ}/s$). Subsequent high-G planar tests confirmed that the raw signal accurately captured peak kinematic velocities of 824$^\circ/s$ without clipping, resolving the truncation drift.

2.2 Iteration II: Network-Induced Loop Instability and Asynchronous Decoupling
Methodology:
With the silicon uncapped, planar testing resumed. However, the integration tracker remained highly erratic, occasionally exhibiting massive, instantaneous jumps in the calculated angle even during slow, controlled movements. To diagnose this, a telemetry hook was added to monitor the Δt variable (the time elapsed between loop cycles) which was theoretically targeted at a stable 200Hz (0.005 seconds).

Findings:
The diagnostic revealed catastrophic processing bottlenecks. The ESP32 microcontroller was executing the kinematic mathematics and the Wi-Fi UDP network transmissions synchronously on the same thread. Furthermore, the telemetry payload was being constructed using computationally heavy String concatenation. Network congestion and router latency caused the loop to occasionally block (freeze) for up to 0.57 seconds.

Because the Riemann sum mathematically assumes a constant velocity over a given time step, a half-second network stutter at the exact moment the arm was moving fast forced the equation to multiply a high velocity by a massive time delta. The algorithm falsely assumed the arm was spinning at maximum speed for the entire duration of the blackout, causing an explosive integration leap that destroyed the coordinate frame.

Results & Resolution:
The firmware architecture was fundamentally rebuilt into an asynchronous, decoupled state machine.

1. **The Math Core:** The kinematic integration was isolated to a non-blocking micros() hardware timer, forcing the math to execute at an ultra-fast, uninterrupted 500Hz (2ms intervals), regardless of network state.
2.
3. **The Telemetry Core:** A secondary, independent timer was established at 60Hz. Heavy String operations were purged and replaced with highly optimized C-style memory buffers (snprintf).
4. This decoupling completely insulated the mathematical integration from Wi-Fi lag spikes, resulting in a perfectly stable, high-resolution Δt baseline.
5.

2.3 Iteration III: Micro-Voltage Noise and State-Machine Filtering
Methodology:

With velocity clipping and network latency resolved, the planar tracker performed excellently during movement but suffered from a slow, constant "phantom rotation" when perfectly stationary. MEMS sensors inherently suffer from thermo-mechanical white noise; even when motionless, the gyroscope outputs micro-voltage fluctuations that mimic a physical rotation of $\sim 0.8^{\circ}/s$. Two distinct software filters were engineered and simultaneously tested to gate this noise: a static Deadband Filter and a dynamic Schmitt Trigger.

Findings:

1. **The Deadband Filter:** A strict threshold boundary was established (e.g., any velocity between $-1.0^{\circ}/s$ and $+1.0^{\circ}/s$ is mathematically forced to exactly zero). While this successfully halted stationary drift, it acted as a blunt "brick wall." As human movements naturally slowed down at the end of an arm swing, the deadband prematurely chopped off the trailing kinetic tail, resulting in a robotic, jagged tracking response that "lost" degrees over time.

2. **The Schmitt Trigger (Hysteresis):** A more sophisticated dual-threshold state machine was designed. It utilized a higher "Wake Threshold" ($1.5^{\circ}/s$) to open the logic gate and a lower "Sustain Threshold" ($0.3^{\circ}/s$) to keep it open.

Results & Resolution:
The Schmitt Trigger proved vastly superior for anatomical tracking. The noise gate remained tightly closed during absolute stillness, successfully zeroing out the stationary micro-voltage noise. When a movement breached the $1.5^{\circ}/s$ wake threshold, the algorithm "latched" the gate open. As the arm smoothly decelerated to a stop, the filter mathematically tracked the velocity all the way down to $0.3^{\circ}/s$ before snapping shut.

This hysteresis logic preserved the delicate trailing edges of human movement. At the conclusion of this planar phase, the uncapped, asynchronous, Schmitt-triggered 1D pipeline achieved an extraordinary error rate of less than 1° over 35 seconds of continuous, bidirectional movement, proving the foundation was mathematically sound and ready to be scaled into three-dimensional space.

3. Three-Dimensional Space: Euler Angles and 6DOF Sensor Fusion

With the fundamental hardware constraints, signal clipping, and network latency resolved on a 2D plane, the methodology progressed to tracking unconstrained three-dimensional anatomical movement. This phase required abandoning simple planar logic in favor of complex multi-axis sensor fusion, ultimately exposing the mathematical limitations of traditional Euler angles.

Geometric Cross-Axis Projection and Planar Failure

The initial 3D trial involved lifting the IMU off the testing surface and moving it through a standard range of motion. The previously optimized 1D Schmitt Trigger filter suffered an immediate and catastrophic failure, accumulating 67.8 degrees of unrecoverable phantom drift.

This failure was caused by geometric cross-axis projection. When an IMU is perfectly flat, global yaw rotates exclusively around the sensor's local Z-axis. However, the moment the human wrist pitches or rolls, the physical rotation vector is divided geometrically across the X, Y, and Z gyroscopes. Because the 1D algorithm exclusively monitored the Z-axis, the physical yaw motion that bled into the X and Y axes was permanently excluded from the integration. This mathematical severing of the coordinate frame proved that single-axis tracking is physically incompatible with anatomical joint movement.

Attitude and Heading Reference Systems (AHRS): Madgwick vs. Mahony
To resolve cross-axis projection, the system required a full 6-Degrees-of-Freedom (6DOF) sensor fusion pipeline. An AHRS algorithm continuously fuses data from the 3-axis accelerometer and the 3-axis gyroscope to compute a unified global gravity vector. Two industry-standard quaternion-to-Euler AHRS architectures were implemented and tested simultaneously:

- **Madgwick Filter:** Employs gradient descent, an analytically derived optimization algorithm, to compute the direction of the gyroscope measurement error as a quaternion derivative. It is computationally demanding but renowned for its smooth response to dynamic motion.

- **Mahony Filter:** Operates as a Proportional-Integral (PI) controller. It utilizes a proportional gain (Kp) to correct immediate vector deviations and an integral gain (Ki) to actively hunt down and cancel sustained micro-drift by tracking the error between the measured and estimated gravity vectors over time.

The Static Bias Calamity and The Integral Gain Trap
Initial uncalibrated 3D trials of both AHRS filters resulted in a disastrous 51-degree drift over a 60-second test. Diagnostic analysis revealed a critical limitation in 6DOF tracking: while the accelerometer permanently locks Pitch and Roll to global gravity (which points straight down along the Z-axis), gravity provides absolutely zero mathematical reference for horizontal Yaw. Consequently, the AHRS yaw tracking relied entirely on the uncalibrated gyroscope, meaning the algorithms flawlessly integrated the hardware's natural static bias ($\sim -0.8^{\circ}/s$) for the entire duration of the test.

To resolve this, a 3-axis dynamic boot calibration sequence was implemented to mathematically zero the hardware bias prior to integration. Following calibration, a violent 76-second 3D stress test was conducted. The Madgwick filter executed flawlessly, maintaining a pristine 1.04 degrees of total drift. Conversely, the Mahony filter accumulated an unacceptable 22.93 degrees of error.

The Mahony failure was diagnosed as an Integral Gain (Ki) trap. The severe linear G-forces generated by rapid human arm swings temporarily overpowered the accelerometer, convincing the sensor that the global "down" vector had violently shifted laterally. The Ki loop aggressively chased these false gravity spikes, permanently corrupting the yaw reference. The Madgwick filter's gradient descent architecture gracefully ignored these high-G linear distortions, cementing it as the optimal tracking engine for wearable robotics.

Nyquist Violations, Euler Wrap-Around, and Gimbal Lock

Despite achieving flawless internal sensor fusion, converting the 4D Madgwick quaternion into a 1D Euler Yaw angle for telemetry introduced severe mathematical boundaries. Euler yaw is strictly bounded to a circle between -180° and $+180^{\circ}$. Rotating past this physical boundary caused the mathematical output to instantly snap to the opposite sign, generating massive $\sim 360^{\circ}$ visual artifacts in the telemetry data.

Attempts to apply an "unwrap" function within the Python telemetry visualizer failed catastrophically during a 395-millisecond Wi-Fi lag spike. The human arm rotated more than 180 degrees during the network blackout. This violated the **Nyquist-Shannon Sampling Theorem**, as the visualizer failed to sample at a frequency greater than twice the highest frequency of the physical rotation. The software observed the pre- and post-blackout angles, incorrectly deduced that the arm had taken the shortest backward path, and applied a permanent -360° phantom offset to the data stream. This was resolved by pushing the continuous unwrap logic down to the ESP32's 500Hz metal loop, forcing the microcontroller to calculate the boundary crossing every 2 milliseconds and rendering the kinematic stream completely immune to network dropouts.

The finalized Euler tracking pipeline faced one ultimate anatomical test: pitching the wrist 90 degrees. At exactly 90 degrees of pitch, the tracker violently disintegrated, registering 1,169 degrees of phantom rotation. This was the manifestation of **Gimbal Lock**. At 90 degrees of elevation, the physical axes for Yaw and Roll merge into a single coplanar dimension. The trigonometric atan2() formula utilized to derive the Euler angles encountered a $0/0$ singularity, causing the coordinate frame to infinitely flip signs. This failure conclusively proved that Euler angles are fundamentally incapable of supporting unconstrained humanoid teleoperation.

4. Phase IV: 4D Quaternion Kinematics, The Digital Twin, and ESP-NOW Scaling

The catastrophic failure of the Euler coordinate system at 90 degrees of elevation necessitated a paradigm shift in how spatial orientation was processed and transmitted. To achieve true 1:1 humanoid teleoperation, the telemetric pipeline had to be mathematically immunized against Gimbal Lock, requiring the permanent abandonment of 3D trigonometric projections in favor of higher-dimensional algebra.

4.1 Eradicating Gimbal Lock via 4D Quaternions

The fundamental flaw of Euler angles (Yaw, Pitch, Roll) is their reliance on sequential, three-axis rotations that are inherently vulnerable to coplanar singularities. To bypass this, the

teleoperation architecture was upgraded to utilize the raw, unadulterated output of the Madgwick filter: **Quaternions**.

A quaternion is a complex mathematical number system operating in four-dimensional space, represented as $[W, X, Y, Z]$. Instead of describing an orientation through a sequence of three sequential rotations, a quaternion represents an orientation as a single rotation around a normalized arbitrary 3D axis.

- **The Mathematical Advantage:** Because quaternions map rotations to points on a 4-dimensional hypersphere rather than projecting them onto a 2D Cartesian plane, they do not rely on trigonometric atan2() functions to determine spatial state. Consequently, it is mathematically impossible for two axes to align and merge.

- **Firmware Execution:** The ESP32 payload was significantly streamlined. The computationally expensive trigonometric Euler conversions and 500Hz unwrapping algorithms were entirely excised from the microcontroller. The payload was reconfigured to transmit the raw, normalized quaternion components directly over the network at 60Hz. Telemetry diagnostics confirmed that the quaternion vector magnitude ($W^2 + X^2 + Y^2 + Z^2$) perfectly maintained a rigid structural normalization of 1.0000 (with an imperceptible standard deviation of 0.000029), guaranteeing the absolute mathematical integrity of the tracking space.

4.2 The Vector-Based 3D Digital Twin

Visualizing 4D hypersphere mathematics on a standard 2D line graph is counterintuitive and practically useless for real-time anatomical tracking. To validate the quaternion pipeline, the Python 2D graphing UI was replaced with a literal 3D digital twin of the human wrist.

Rather than converting the quaternion back into flawed Euler angles for rendering, the 3D visualizer was engineered to extract explicit directional vectors directly from the raw W, X, Y, Z data. By applying specialized rotation matrices, the system continuously calculates the local "Forward" (x-axis) and "Up" (z-axis) physical vectors of the IMU:

Forward Vector (f_x, f_y, f_z):

- $f_x = 1 - 2(Y^2 + Z^2)$
- $f_y = 2(XY + WZ)$
- $f_z = 2(XZ - WY)$

Up Vector (u_x, u_y, u_z):

- $u_x = 2(XZ + WY)$
- $u_y = 2(YZ - WX)$

- $u_z = 1 - 2(X^2 + Y^2)$

These derived Cartesian vectors dictate the absolute spatial orientation of a rendered 3D VPython mesh. The resulting digital twin acts as a flawless 1:1 mirror of the human anatomy. Regardless of extreme spatial contortion, rapid 90-degree wrist pitches, or violent high-velocity swings, the 3D twin remains locked to reality—completely invulnerable to Gimbal Lock, singularity flips, or Euler wrap-around tearing.

4.3 Multi-Node Scaling via ESP-NOW Protocol

With the fundamental kinematic mathematics and firmware architectures perfected on a single, isolated sensor node, the framework was finalized for mass scaling. A functional upper-body VR teleoperation harness requires five independent IMU nodes (e.g., Left/Right Hands, Left/Right Forearms, and Torso/Head) transmitting simultaneously.

Relying on standard Wi-Fi UDP for five concurrent datastreams introduces severe router bottlenecking, packet collision, and unacceptable dynamic latency for real-time robotics. To resolve this, the network architecture is transitioned to the **ESP-NOW** protocol.

- **MAC-Layer Telemetry:** ESP-NOW operates directly on the Media Access Control (MAC) layer, completely bypassing the massive overhead of standard Wi-Fi router protocols (TCP/IP stack, DHCP, and router hopping).

- **Deterministic Latency:** By enabling peer-to-peer, connectionless broadcasting, the 5 independent transmitter nodes can instantly blast their 4D quaternions to a central ESP32 receiver wired directly to the Unitree G1. This sub-millisecond, low-latency mesh ensures that the humanoid's inverse kinematics engine receives perfectly synchronized upper-body coordinate frames.

Conclusion

The successful engineering of this 5-node wearable IMU harness represents a critical advancement in accessible humanoid teleoperation. By methodically identifying and eliminating the hardware, network, and mathematical failure points inherent to 6-axis MEMS sensors, this architecture neutralizes the necessity for magnetometer-based 9-axis tracking. Through the synthesis of hardware-flushed DSP initialization, asynchronous 500Hz loop execution, and continuous 4D quaternion sensor fusion, this system successfully bridges the gap between chaotic human kinematics and pristine robotic control.